The World Wide Web

Competency 10: Develops websites incorporating multi-media technologies (using HTML 5)

Competency Level 10.1: Explores the need for web

Learning Outcomes:

- Describes www
- Analyses the systematic arrangements of contents and structure of a web

The World Wide Web (WWW) invented by Tim Berners Lee in 1989 at the European Centre for Nuclear Research, as a major application on the Internet. It was the first application on the Internet that caught the attention of the public and it dramatically changed how people interact with the network of computers. Essentially, the World Wide Web (WWW) is an application that runs on top of the Internet, which is in turn made up of a lot of interconnected computers. According to Wikipedia, World Wide Web is an information system where documents and other web resources are identified by Uniform Resource Locators(URLs), which may may be interlinked by hypertext, and are accessible over the Internet. The basic element in WWW is the web page that a user views when he or she accesses it. Some of the computers on the Internet, called web servers, are designed to serve the web pages. These web servers are always on computers and are ready to serve the web pages all the time. The other computers request web pages from the web servers and are called the client computers. The WWW operates on-demand, the users request web pages from web servers whenever they want and receive them from web servers. The WWW has many wonderful features that people like a lot; it provides a platform for any individual who has access to a computer and the Internet to make information available on the web at very less cost. Formatted Text, multimedia, graphics, forms, and applets are part of WWW and are accessible by the client computers.

WWW also serves as a platform for many innovative applications like search engines such as Google, other popular Google applications such as e-mail, Social media such as Facebook, multimedia streaming platforms such as YouTube, online photo sharing applications such as Flickr, etc.

A Web page can consist of a number of web objects such as a simple text file, for example, an HTML file, an image file such as a jpeg file, a Java applet, an audio file or a video clip. The text in a web page called hypertext means that it is the text, which contains links to other text. Any word in a hypertext document can be specified as a pointer to a different hypertext document where more information pertaining to that word can be found. A web page stored in a server computer can be accessed by issuing the address of the server, which houses the resource or object, called a Uniform Resource Locator (URL), or a hyperlink. Each URL has three components: the first part is the protocol, which tells the browser how to communicate with a website's server, in order to send and retrieve information, the second part is the domain name, which is an identifier to a specific site, and the third part is the path, which directs the browser to a specific page on the website. Typically a web page will consist of a base HTML file and

several other objects that are embedded into the base HTML file. Each embedded object will be referred to using its URL. If a given web page has n number of other objects referred in it, then we have a total of (n+1) objects on the web page including the base HTML file.

Web pages requested by clients can be viewed using a special application called web browsers. Present-day web browsers are intelligent applications and they provide many features to the users. Google Chrome, Mozilla Firefox, Internet Explorer and Apple's Safari are examples of popular web browsers. Web pages are written in HyperText Markup Language (HTML). The markup facility of the HTML enables the web browsers to render plain text, combined with additional tags, as easily readable attractive text on the display of computers. HTML Tags are symbols that indicate a visual element for the page, such as a heading, that governs the size, color, and other properties of the element. Each of these web browser applications have their strengths and weaknesses, and can render the HTML code slightly differently. It would be better that when HTML code is written it should be tested on each of this web browser applicaton software for completeness.

Figure 3.1 some of the well-known web browsers

Competency Level 10.2: Analyses user requirements (multimedia contents)

Learning Outcomes:

- Creates effective and appropriate information layout of a website
- Identifies the web pages of a website
- Identifies the contents of a web page

Identifies navigation structure

Analyses user requirements (multimedia contents)

Objectives of a web site should be defined well to build a great web site. It helps the designer to identify the requirements of the users, to define them and to satisfy the user requirements. A set of well defined such objectives and their usability solutions can increase the user experience and will keep the users regularly visit the web sites.

Some possible questions about the objectives of a website are:

1. Why do you need a new website?
2. What outcomes do you want to achieve?
3. Do you have an existing website? Are there any problems with it?
4. How do you plan to improve the user experience? E.g. Speed, content, clarity, etc
5. What do you try to boost through the new website?
6. Are you going to inform about new opportunities available in your organization?
7. Are you planning to expand your business?

You have to define some goals of having a website that helps you to achive your objectives. For example, increasing the number of inquiries to the website or increasing online sales can be defined as goals for an online shopping website.

You may also want to identify the key audiences that you need to appeal to in order to reach your objectives. The set of your key audiences may include the potential customers of your business and your employees. Once you have identified the prospective audiences, it is better to list the priorities of these audiences and doing so will help you in planning the design layout and the navigation design of your website.

Requirements Gathering for a Website

User requirements of a web site describe the needs of users, the functionalities that the web site should do, and the goals of a web site. The designer of the website should define, document and describe the user requirements of the web site. These requirements would change over time and the designers should keep the website updated based on the changing requirements. Some of the possible requirements that may be collected before commencing the design are listed below:

- Content – Determining the content of the web site that needs to be updated on the web site is an essential part of the design.
- Layout of the design and the navigation pattern – Determining what would appear in the layout of the website, and how the user would be able to navigate from the home page to other pages.
- Usability – identifying the accessibility requirements of the web site.

- Security – identifying the security requirements of the web site.
- Loading times – determining the loading speed of a website depending on the kind of service it provides.
- Legal – identifying the legal implications that the web site must adhere to.

Competency Level 10.3: Identifies appropriate HTML tags to design a single web page

Learning Outcomes:

- Analyses the arrangement of contents of a web page
- Analyses the organization of contents in a web page
- Creates a simple web page

Identifies appropriate HTML tags to design a single web page

HTML Tags are symbols that indicate a visual element for the page, such as a heading, that governs the size, color, and other properties of the element. When a web browser loads a web page, it renders the content of the web page on the computer's display according to the included HTML markup and content.

A number of tools are available to create HTML pages and for learning purposes, the notepad application available in desktop computer is sufficient. Notepad is a lightweight application that enables the typing of HTML pages without much trouble. Typed web pages could be saved using the extension .html. There exist multiple sophisticated tools that enable HTML typing and there even other online resources available to type in HTML.

The structure of an HTML page

An HTML page contains a series of tags; the larger tag is the <html> tag that contains two structural elements <head> and <body>. The essential code of an HTML5 looks like as follows:

Code snippet:

```
<!DOCTYPE html>
<html>
    <head>
    </head>
    <body>
    </body>

</html>
```

Output:

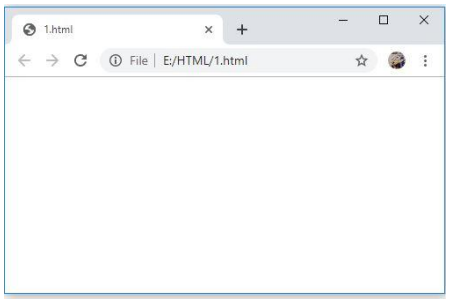

Figure 3.3.1

Figure 3.3.1 First example of html code to create an empty webpage.

The above document does not produce any output since it does not have any content to display. The <! DOCTYPEHTML> tells the browser what type of document is following the file. When the browser knows the type of the document, it can present it to the viewer of the document in a much faster manner.

The primary tag of a web page is the <html> tag. All the content that is presented to the browser for rendering should be included within the <html> and </html> pair.

Inside the <html> and </html> tags are two main components, the <head> and the <body>. The head section contains information about the current document, often referred to as metadata. This metadata may include the title of the document, keywords and descriptions that describe the page, author details, and copyright statements among other information. The <title> and </title> pair within the head are used to inform the title of the webpage which is displayed at the topmost bar of the browser.

The <body> and</body> contains all the content of the web page that can be rendered through the browser. Within the body pair, the code that can be used to style the web page is written. By including these two major sections of a web page, the structure of a web page will look like:

```
<!DOCTYPE html>
<html>
    <head>
            <title> Title of the document which appears at the title bar of the browser
    </title>
    </head>
    <body>
        The content of the document appears here
    </body>
</html>
```

Output:

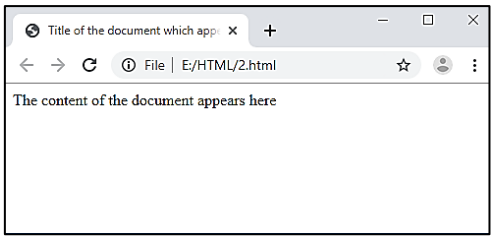

Figure 3.3.2

The <body> tag is capable of accepting numerous attributes, including the ID attribute, for example, <body id="home">. A page with a distinctive ID in the <body> tag can be targeted for specific styling using CSS. We will study more about the other possible attribute when we study about the CSS.

Typing in an HTML document

To type in an HTML document you can use a simple editor such as notepad, and to view the web page that you have created you may use a web browser such as Mozilla Firefox, Google Chrome or Apples Safari. You can type in the code as shown above in the text editor and then you can save the HTML document with a name using the .html extension, for example, html_example1.html. Now you may notice that the saved document has an icon of the browser that you use on your computer. The saved web document can now be viewed using a web browser, by double-clicking on the icon of the document and the output will look like:

Figure 3.3.3

Background colour in HTML:
A web page can be assigned a background colour of choice by adding an extra attribute to the <body> tag.

E.g. <body bgcolor="#E6E6FA"> can be used to assign a background colour to the html document.Syntax of it is <body bgcolor="color_name|hex_number|rgb_number">, where color_name can specify a particular colour such as “Blue”, hex_number can specify the hexa decimal code of a colour, and rgb_number can specify the RGB code.
Note: This facility for setting background colour is available in previous versions of HTML and, in HTML 5 these will come under CSS. We will cover them when we study CSS later in this chapter.

Heading tags in HTML:
Headings are used to separate and introduce major divisions within a web page. The <h1> to <h6> tags in HTML are used to define the six levels of headings on a web page according to their importance. <h1> defines the most important heading whereas the <h6> defines the least important heading.

Formatting text in HTML:

The .. pair defines paragraph in HTML and the text within the .. pair are displayed in the plain text format. In HTML, the tag <br> is used to insert a single line break. It can be used to break sentences into a number of separate lines. It is referred to as an empty tag since it does not have an ending tag. The .. tag pair is used to **bold** the text that is written within the pair. The <u> .. </u> tag pair is used to <u>underline</u> a given sentence written within the pair. The .. tag pair is used to define the part of a text in *italics*. The <font>.. </font> pair is used to specify the font to be used in the text. It also allows specifying the font size and colour of the text. In between paragraphs of text, a <br> tag can be inserted to include a single line break in the text. The <br> tag is an empty tag and it does not have an end tag.

Note: At times developers of web pages need to include comments in their web pages for maintenance purposes. These comments help the developers to debug the code in case of errors in rendering the web pages. Comments within HTML code can be included using the <!-- and --> tag pair.

Let us see an example of the above formatting tags in an html document:

Example Code:

```
<!DOCTYPE html>
<html>
    <head>
        <title> My new Webpage </title>
    </head>
    <body>
        <h1> Welcome to My New Webpage </h1>.
        <p> This is a paragraph of text.
        </p>
        <p>This is normal text - <b> and this is bold text </b></p>
        <p>This is normal text - <u> and this is underlined text </u></p>
        <p>He named his car <i>The lightning</i>, because it was very fast.</p>
        <font size="3" color="red"> This is some text! </font>
        <font size="2" color="blue"> This is some text! </font>
        <font face="verdana" color="green"> This is some text! </font>
        <!-- This is a comment -->
    </body>
</html>
```

When the above code is viewed using a web browser the output will look like:

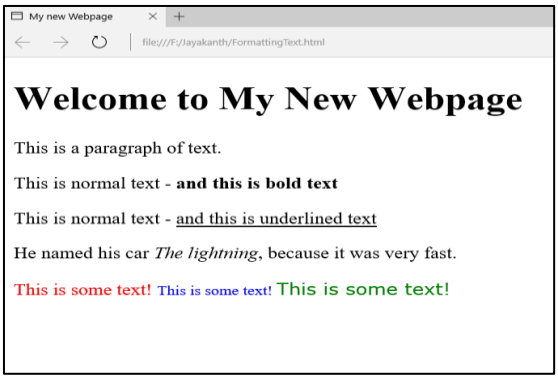

Figure 3.3.4

Fig 3.3.4 html code to display and the corresponding output

Note: There are many other types of text formatting tags available in HTML. Students are encouraged to find them and try them by including them in the above code snippet by doing a self-search on the web. A good source for many of the HTML tags and attributes is w3school.com

Lists in HTML:

Lists are a common text element on the Web, often used to break up the page and highlight key points. When the list items do not need to be in any particular order, an unordered listis used.

Ordered Lists:

Ordered lists are mainly used when the numbering of the items in the list is important. Ordered lists can be created using the <ol>..</ol> tag pair. There are many numbering styles available in the ordered list.

Example:

```
<ol>
        <li> First item </li>
        <li> Second item </li>
        <li> Third item </li>
        <li> Fourth item </li>
</ol>
```

Note: The outer wrapping tag <ol> can have a number of attributes added to it. The *start* attribute in the <ol> tag can be used to specify a number to start the ordered list to begin numbering than the usual number "1". For example,
<ol start = "100"> will enable the list to begin with a number 100. In this case the first item would be numbered as 100, second would be numbered with 101, and third would be 102, and so on.

Nested ordered lists

Ordered lists can be nested therefore they can multiple levels of numbering within a listing. For Example, let us create a nested ordered list as follows:

```
<ol>
        <li>Installation
</ol>
                        <li>Computer set up</li>
                        <li>Monitor set up
                <ol>
                        <li>Model XYZ</li>
                        <li>Model ABC</li>
                </ol>
        </li>

        </ol>
        </li>
        <li>Maintenance</li>
        <li>Use</li>
</ol>
```

When viewed using a browser, the above code snippet will generate an output as follows:

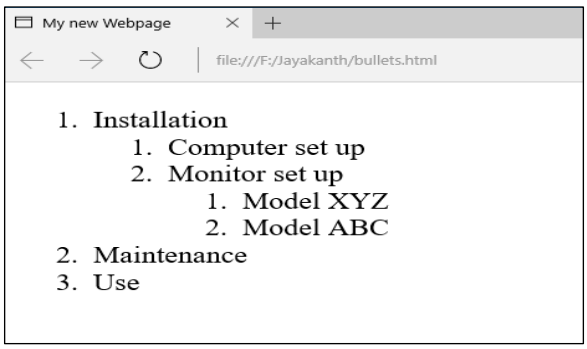

Figure 3.3.5

The above output shows the different levels with the restarting of numbers. You can change the numbering from the standard Decimal numbers to different styles using CSS. We will see it later when we study the Cascading Style Sheets.

Note: Unordered lists and ordered lists can be combined in any sequence by nesting one within the other.

Unordered List:
In HTML, an unordered list is composed of two tags: <ul> and <li>. The <ul> tag is the outermost structure that declares the unordered list. Within the <ul> tag, a series of <li> tags creates the items in the list. Here's a short example of the HTML code for an unordered list:

(write about the bullets in lists and there are many different styles to include)

```
<ul>
        <li>Tomatoes</li>
        <li>Onion</li>
        <li>Garlic</li>
</ul>
```

Figure 3.3.6

Note: The <li> tag can contain any amount of text, from a single word to multiple lines of text. Bullets in a list can be a standard unordered list gives equal weight to all the bulleted items, one after another. In some situations, it is desirable to depict multiple levels of content with sub-items. HTML provides the capacity to incorporate any level of sub-items desired by nesting <ul> tags.

Like unordered lists, the numbered variety uses two key elements: an outer wrapping tag and a separate tag for each list item. The only difference is that the outer tag is not <ul> but ol>. Here's a brief example:

```
<ol>
        <li>Pull mask from overhead bin</li>
        <li>Place mask over face</li>
        <li>Pull strings tight</li>
</ol>
```

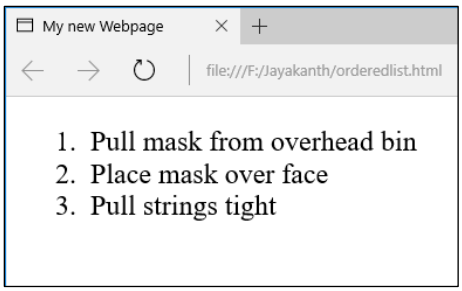

Figure 3.3.7

Table

In html you can create a table using the three different tags: <table>: This is the outermost element that contains the other two tags and all content. <tr>: The <tr> tag defines the table row and holds the final element. <td>: The <td> tag stands for table data; the complete <td> tag is also known as a table cell. Any content that is displayed in the table is placed between the opening and closing <td> tag pair. For example,

```
<table>
        <tr>
                <td>First name</td>
                <td>Last name</td>
                <td>Extension</td>
        </tr>
        <tr>
                <td>Pat</td>
                <td>Peterson</td>
                <td>x394</td>
        </tr>
</table>
```

It will create a table that has three columns and two rows.

When viewed using a browser, the above code snippet will generate an output as follows:

Figure 3.3.8

The above declaration does not have column header and a column header can be easily added by having a <th> tag instead of a <td> tag. For example,

```
<table><tr>
                <th>First name</th>
                <th>Last name</th>
                <th>Extension</th>
        </tr>
        <tr>
                <td>Pat</td>
                <td>Peterson</td>
                <td>x394</td>
        </tr>
</table>
```

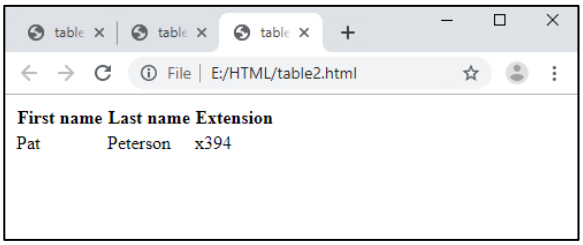

Figure 3.3.9

In this case we have explicitly said the table headers and the table data. Browsers will usually display the header of the table in bold letter. HTML has more tags that enable the creation of more structured tables with header, body and footer regions. For example,

```
<table>
        <thead>
                <tr>
                        <th>Region</th>
                        <th>Sales</th>
                        <th>Amount</th>
                </tr>
        </thead>
        <tfoot>
                <tr>
                        <th>Total</th>
                        <th> </th>
                        <th>$6,500</th>
                </tr>
        </tfoot>
        <tbody>
                <tr>
                        <td>North</td>
                        <td>Peterson</td>
                        <td></td>
                </tr>
                <tr>
                        <td>South</td>
                        <td>Kattrell</td>
                        <td>x396</td>
                </tr>
        </tbody>
</table>
```

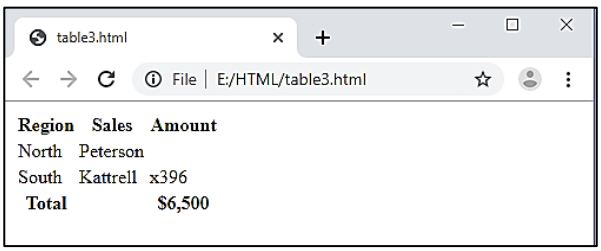

Figure 3.3.10

The colspan and rowspan attributes both take numeric values to define how many columns or rows will be spanned, respectively. For example, if I wanted to create a table that had two headers, each of which spanned two of the four columns, my code would look like this:

```
<table border =2>
        <tr>
                <th colspan="2">Atlantic Division</th>
                <th colspan="2">Pacific Division</th>
        </tr>
        <tr>
                <td>New York</td>
                <td>Boston</td>
                <td>San Francisco</td>
                <td>Los Angeles</td>
        </tr>
</table>
```

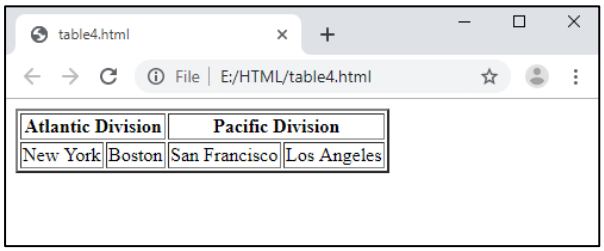

Figure 3.3.11

When rendered in the browser, the headers are centered over the spanned columns as shown in Figure 3.3.11.To better show the spanning and centering, I added a CSS rule to give the table a width of 300 pixels as well as another to create the outlining borders. Implementing the rowspan attribute requires a different table configuration than what is needed for colspan:

```
<table border=2>
        <tr>
        <th rowspan=2>Atlantic Division</th>
        <td>New York</td>
        </tr>
        <tr>
        <td>Boston</td>
        </tr>
        <tr>
        <th rowspan=2>Pacific Division</th>
        <td>San Francisco</td>
        </tr>
        <tr>
        <td>Los Angeles</td>
        </tr>
</table>
```

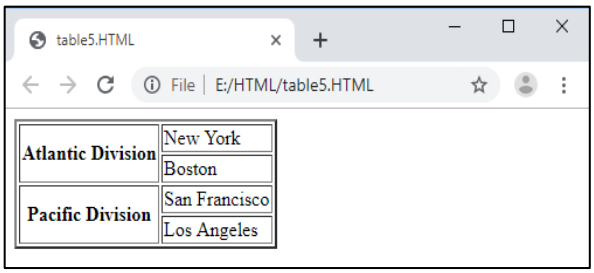

Figure 3.3.12

Competency Level 10.4: Uses HTML to create linked web pages

Learning Outcomes:

- Explains hypertext markup language
- Identifies the standards of HTML
- Saves the source document with suitable extensions
- Designs the web page by inserting appropriate multimedia objects according to user requirements
- Organizes data using lists and tables in the web page
- Links pages and multimedia objects to the web page

HTML to create linked web pages

We have looked at the basic structure of a web page that can be written using HTML, and a few other tags that can be included to create a web page. We have also seen that a designed web page can be viewed using a web browser to view the content of the web page. Now let us see how we create a set of linked web pages and their properties.

Html is the language of the web. As we know HTML stands for a hypertext markup language. In writing web pages, we have hypertext which implies the text is not normal text but is hypertext that has a link which when clicked by a mouse pointer can take the user to a new web page or to another portion of the text in the same web page. The markup language component allows us to mark up our programs with additional tags and other attributes to further enhance the text. Therefore, they could be more readable on the displays. HTML5 is the latest version of HTML and it has many advanced features such as the ability to play built-in videos and animations without the need to download additional add on, than the previous versions.

A web page can contain text and audiovisual content on it and can be presented to the user as a web page. Having just one web page will not serve the purpose for many enterprises and there need to be a number of interconnected web pages designed to present the content to the viewers. Since enterprises have much more information that needs to be classified and presented differently, multiple web pages that present the information in such a way are needed to be developed and linked together. When a number of web pages are developed and linked together it is called a web site. A website for an enterprise contains a number of interconnected web pages that contain hyperlinks, therefore pages could be navigated from one to another. The main web page of a web site is called the home page and it is assumed to be the starting page of a web site from which the user begins navigating the web site. The home page also facilitates the navigation to other web pages of the web site by providing navigational links in it.

Linked pages of a website

Hyperlinks

InanHTML document, there will be a number of links that have a connection to a content that appears on the same web page, or a connection that takes you to a new webpage. Such links in an HTML document are called hyperlinks. Web pages can be linked to each other therefore the web pages can be navigated back and forth easily and the user experience can be improved.

To jump from one page to another, the <a> tag, the anchor tag can be used. The text or image enclosed by the <a> tag anchors one side of the link to the current page, and the href attribute (short for hypertext reference) specifies the other side, the destination. For example, consider,

```
<a href=”home.html”>Home</a>
```